

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I K. Yamini (192572171) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Yamini

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20
22-11

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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COURSE NAME: COMPUTER NETWORKS

SCENARIO-BASED

SCENARIO 1: SMART CAMPUS NETWORK DESIGN

1. Understanding of Scenario

The IP address block 192.168.100.0/24 is subnetted using Variable Length Subnet Masking (VLSM) to divide the network into smaller subnets based on varying departmental requirements. This ensures efficient allocation of IP addresses while minimizing wastage.

2. Identification of Key Issues

- Requirement of different subnet sizes for each department
- Efficient use of available IP address space
- Accurate calculation of network and broadcast addresses
- Determination of valid host ranges for communication
- Avoidance of overlapping IP ranges
- Planning for scalability and future expansion

3. Application of Concepts

3.1 School of Computing (/25)

Network Address: 192.168.100.0

Broadcast Address: 192.168.100.127

Host Range: 192.168.100.1 – 192.168.100.126

Usable Hosts: 126

Explanation: A /25 subnet provides 128 addresses, out of which 126 are usable for hosts, making it suitable for large departments.

3.2 School of Electronics (/26)

Network Address: 192.168.100.128

Broadcast Address: 192.168.100.191

Host Range: 192.168.100.129 – 192.168.100.190

Usable Hosts: 62

Explanation: A /26 subnet provides 64 addresses with 62 usable hosts, ideal for medium-sized departments.

3.3 School of Mechanical Engineering (/27)

Network Address: 192.168.100.192

Broadcast Address: 192.168.100.223

Host Range: 192.168.100.193 – 192.168.100.222

Usable Hosts: 30

Explanation: A /27 subnet provides 32 addresses with 30 usable hosts, suitable for smaller departments.

4. Decision Making

Subnets are allocated in descending order of size (largest to smallest) to optimize IP utilization, prevent wastage, and allow future scalability. This structured approach ensures efficient network management.

SCENARIO 2: COMPANY NETWORK DESIGN

1. Understanding of Scenario

The network 10.10.0.0/16 is divided into multiple subnets according to departmental needs, ensuring logical segmentation and efficient management of network resources.

2. Identification of Key Issues

- Allocation of subnets based on number of required hosts
- Efficient division of a large IP block
- Identification of network and broadcast addresses
- Ensuring no overlap between subnets
- Planning hierarchical network structure

3. Application of Concepts

3.1 Software Development (/24)

Network Address: 10.10.0.0

Broadcast Address: 10.10.0.255

Host Range: 10.10.0.1 – 10.10.0.254

Usable Hosts: 254

Explanation: A /24 subnet supports large teams and provides sufficient host capacity.

3.2 Quality Assurance (/25)

Network Address: 10.10.1.0

Broadcast Address: 10.10.1.127

Host Range: 10.10.1.1 – 10.10.1.126

Usable Hosts: 126

Explanation: Suitable for medium-sized teams requiring moderate host capacity.

3.3 Human Resources (/26)

Network Address: 10.10.1.128

Broadcast Address: 10.10.1.191

Host Range: 10.10.1.129 – 10.10.1.190

Usable Hosts: 62

Explanation: Designed for smaller departments with fewer devices.

3.4 Executive Management (/27)

Network Address: 10.10.1.192

Broadcast Address: 10.10.1.223

Host Range: 10.10.1.193 – 10.10.1.222

Usable Hosts: 30

Explanation: Provides a secure and limited network for high-level management.

4. Decision Making

Subnetting is performed based on organizational hierarchy and department size, ensuring efficient address allocation, improved performance, and simplified network administration.

SCENARIO 3: IPV6 NETWORK ANALYSIS

1. Understanding of Scenario

The IPv6 address is analyzed to understand its structure, convert between compressed and expanded formats, and evaluate its addressing capacity.

2. Identification of Key Issues

- Conversion between compressed and expanded IPv6 formats
- Identification of network prefix and interface ID
- Understanding the massive host capacity of IPv6
- Ensuring proper representation of address structure

3. Application of Concepts

Compressed Form: 2001:db8:abcd::/64

Expanded Form: 2001:0db8:abcd:0000:0000:0000:0000

Network Prefix: First 64 bits (2001:0db8:abcd:0000)

Total Hosts: 2^{64}

Explanation: IPv6 uses 128-bit addressing, and a /64 subnet provides an extremely large number of unique addresses, eliminating address exhaustion issues.

4. Decision Making

IPv6 is implemented to support future network growth, ensure global uniqueness of addresses, and overcome limitations of IPv4.

SCENARIO 4: ROUTING TABLE ANALYSIS

1. Understanding of Scenario

A router determines the path of a packet by comparing the destination IP address with entries in its routing table.

2. Identification of Key Issues

- Matching destination IP with correct subnet
- Use of longest prefix matching
- Identifying correct next-hop or interface
- Handling unmatched routes using default routing

3. Application of Concepts

Destination Match: 192.168.2.0/24

Selected Route: Entry with longest prefix match

Forwarding Action: Packet is sent through the corresponding interface

Default Route: 0.0.0.0/0 (used when no match exists)

Explanation: The router prioritizes the most specific route to ensure accurate and efficient delivery of packets.

4. Decision Making

Using longest prefix matching improves routing efficiency and ensures packets reach the correct destination with minimal delay.

SCENARIO 5: TWO LAN NETWORK CONFIGURATION

1. Understanding of Scenario

Two LAN networks are created using /26 subnetting and connected via a Layer-3 router, enabling communication between different departments.

2. Identification of Key Issues

- Subnet division and address allocation
- Determining network and broadcast addresses
- Assigning valid host IPs
- Configuring default gateways for communication

3. Application of Concepts

3.1 Administration LAN (/26)

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Host Range: 192.168.10.1 – 192.168.10.62

Sample IP Addresses:

192.168.10.2

192.168.10.3

192.168.10.4

Default Gateway: 192.168.10.1

Explanation: The gateway acts as the access point for communication outside the LAN.

3.2 Finance LAN (/26)

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Host Range: 192.168.10.65 – 192.168.10.126

Sample IP Addresses:

192.168.10.66

192.168.10.67

192.168.10.68

Default Gateway: 192.168.10.65

Explanation: Proper gateway configuration ensures successful routing between both LANs.

4. Decision Making

Accurate subnetting and gateway configuration enable efficient communication, reduce network conflicts, and ensure smooth data transfer between departments.